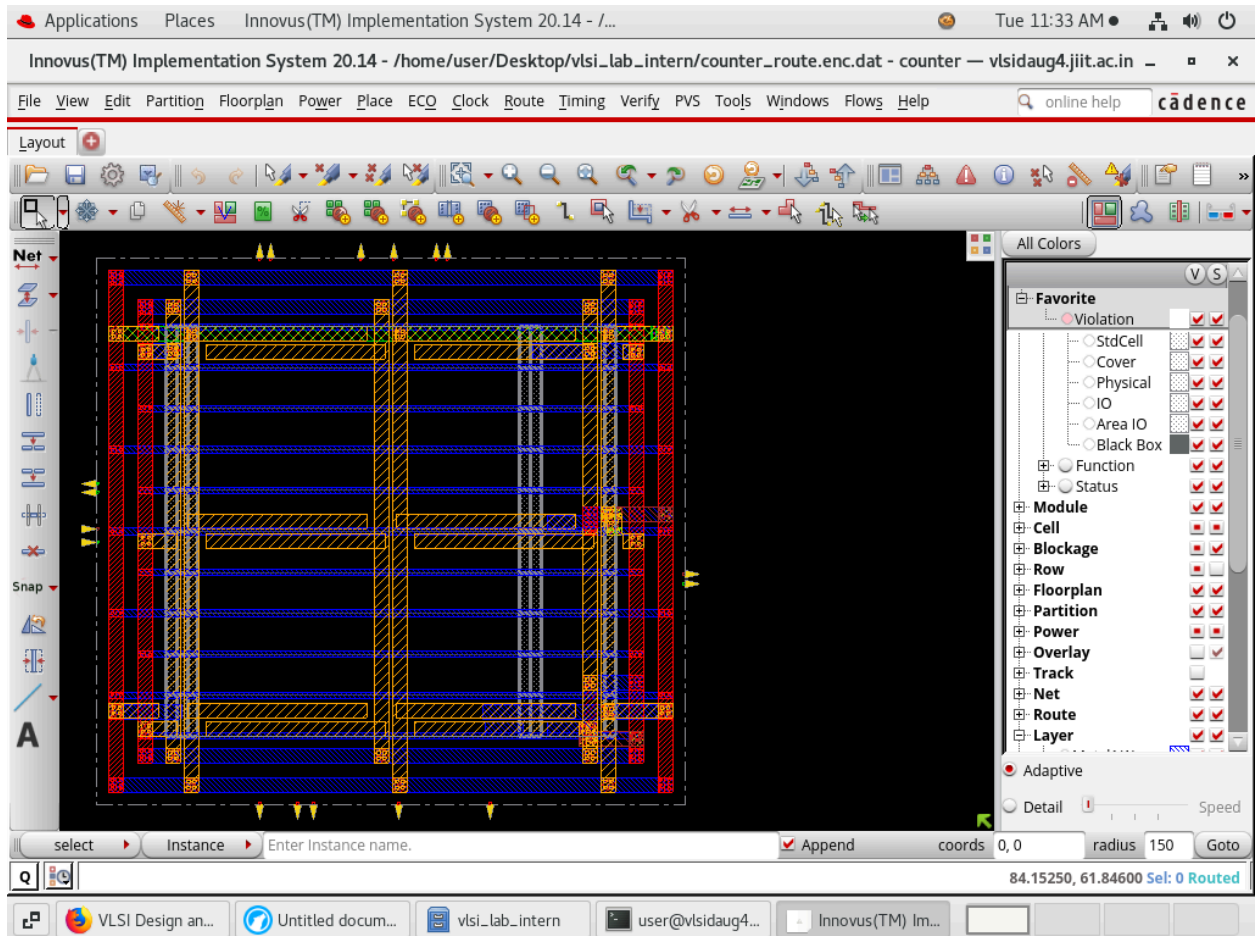


Week-4

1. Counter



Area report

```
innovus 1> report_area
```

Hinst Name	Module Name	Inst Count	Total Area
counter		57	2270.660

Timing report

```
innovus 3> *** timeDesign #2 [begin] : totSession cpu/real = 0:02:27.3/0:28:38.8 (0.1),
mem = 1565.4M
```

Start to check current routing status for nets...

All nets are already routed correctly.

End to check current routing status for nets (mem=1520.4M)

timeDesign Summary

Setup views included:

worst

Setup mode	all	reg2reg	default
WNS (ns):	0.000	N/A	0.000
TNS (ns):	0.000	N/A	0.000
Violating Paths:	0	N/A	0
All Paths:	0	N/A	0

	Real	Total
DRVs	Nr nets(terms)	Worst Vio
max_cap	0 (0)	0.000
max_tran	0 (0)	0.000
max_fanout	0 (0)	0
max_length	0 (0)	0

Density: 93.194%

Routing Overflow: 0.00% H and 0.00% V

Reported timing to dir timingReports

Total CPU time: 0.05 sec

Total Real time: 0.0 sec

Total Memory Usage: 1552.433594 Mbytes

*** timeDesign #2 [finish] : cpu/real = 0:00:00.1/0:00:00.6 (0.1), totSession cpu/real = 0:02:27.4/0:28:39.4 (0.1), mem = 1552.4M

Power report

Power Net Detected:

Voltage	Name
0V	VSS
0.9V	VDD

Using Power View: worst.

Set Default Frequency 100MHz.

Begin Power Analysis

0V VSS

0.9V VDD

Warning:

There are 2 power/gnd nets that are not connected

VSS VDD ...

Use 'globalNetConnect' to define rail connections.

** WARN: (VOLTUS_POWR-2035): There are 2 power/gnd nets that are not connected

VSS VDD ...

Use 'globalNetConnect' to define rail connections.

Begin Processing Timing Library for Power Calculation

Begin Processing Timing Library for Power Calculation

Begin Processing Power Net/Grid for Power Calculation

Ended Processing Power Net/Grid for Power Calculation: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1170.84MB/2712.70MB/1170.87MB)

Begin Processing Timing Window Data for Power Calculation

Ended Processing Timing Window Data for Power Calculation: (cpu=0:00:00,
real=0:00:00, mem(process/total/peak)=1170.96MB/2712.70MB/1170.96MB)

Begin Processing User Attributes

Ended Processing User Attributes: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1171.02MB/2712.70MB/1171.02MB)

Begin Processing Signal Activity

Starting Levelizing

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT)

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 10%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 20%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 30%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 40%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 50%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 60%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 70%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 80%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 90%

Finished Levelizing

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT)

Starting Activity Propagation

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT)

** INFO: (VOLTUS_POWER-1356): No default input activity has been set. Defaulting to 0.2.

Use 'set_default_switching_activity -input_activity' command to change the default activity value.

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 10%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 20%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 30%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 40%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 50%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 60%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 70%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 80%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 90%

Finished Activity Propagation

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT)

Ended Processing Signal Activity: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=1171.27MB/2712.70MB/1171.27MB)

Begin Power Computation

of cell(s) missing both power/leakage table: 0

of cell(s) missing power table: 0

of cell(s) missing leakage table: 0

of MSMV cell(s) missing power_level: 0

Starting Calculating power

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT)

... Calculating switching power

Cannot locate supply power rail for net 'count[15]' of instance

\count_reg[15]

Cannot locate supply power rail for net 'count[14]' of instance

\count_reg[14]

Cannot locate supply power rail for net 'count[13]' of instance

\count_reg[13]

Cannot locate supply power rail for net 'count[12]' of instance

\count_reg[12]

Cannot locate supply power rail for net 'count[11]' of instance

\count_reg[11]

only first five unconnected nets are listed...

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 10%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 20%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 30%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 40%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 50%

... Calculating internal and leakage power

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 60%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 70%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 80%

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT): 90%

Finished Calculating power

2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT)

Ended Power Computation: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1171.71MB/2712.70MB/1171.71MB)

Begin Processing User Attributes

Ended Processing User Attributes: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1171.71MB/2712.70MB/1171.78MB)

Ended Power Analysis: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1171.78MB/2712.70MB/1171.78MB)

Begin Boundary Leakage Calculation

Ended Boundary Leakage Calculation: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1171.79MB/2712.70MB/1171.79MB)

Begin Static Power Report Generation

*-----

* Innovus 20.14-s095_1 (64bit) 04/19/2021 14:41 (Linux 2.6.32-431.11.2.el6.x86_64)

*

*

* Date & Time: 2025-Jun-24 11:38:37 (2025-Jun-24 06:08:37 GMT)

*

```
*-----  
*  
* Design: counter  
*  
* Liberty Libraries used:  
*   worst:  
/home/user/Desktop/vlsi_lab_intern/counter_route.enc.dat/libs/lib/typ/slow.lib  
*  
* Parasitic Files used:  
*  
*   Power View : worst  
*  
*   User-Defined Activity : N.A.  
*  
*   Activity File: N.A.  
*  
*   Hierarchical Global Activity: N.A.  
*  
*   Global Activity: N.A.  
*  
*   Sequential Element Activity: N.A.  
*  
*   Primary Input Activity: 0.200000  
*  
*   Default icg ratio: N.A.  
*  
*   Global Comb ClockGate Ratio: N.A.  
*  
* Power Units = 1mW  
*  
* Time Units = 1e-09 secs  
*  
*   report_power  
*  
*-----  
*
```

Total Power

```
-----  
Total Internal Power:    0.00336494    82.7375%  
Total Switching Power:  0.00068916    16.9452%  
Total Leakage Power:    0.00001290     0.3173%
```

Total Power: 0.00406701

Group	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
Sequential	0.00301	0.0004201	9.612e-06	0.00344	84.58
Macro	0	0	0	0	
IO	0	0	0	0	
Combinational	0.0003548	0.000269	3.291e-06	0.0006271	15.42
Clock (Combinational)		0	0	0	0
Clock (Sequential)		0	0	0	0
Total	0.003365	0.0006892	1.29e-05	0.004067	100

Rail	Voltage	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
Default	0.9	0.003365	0.0006892	1.29e-05	0.004067	100

* Power Distribution Summary:

* Highest Average Power: count_reg[1] (SDFFRHQX8): 0.0004779

* Highest Leakage Power: count_reg[1] (SDFFRHQX8): 7.563e-07

* Total Cap: 3.79519e-13 F

* Total instances in design: 57

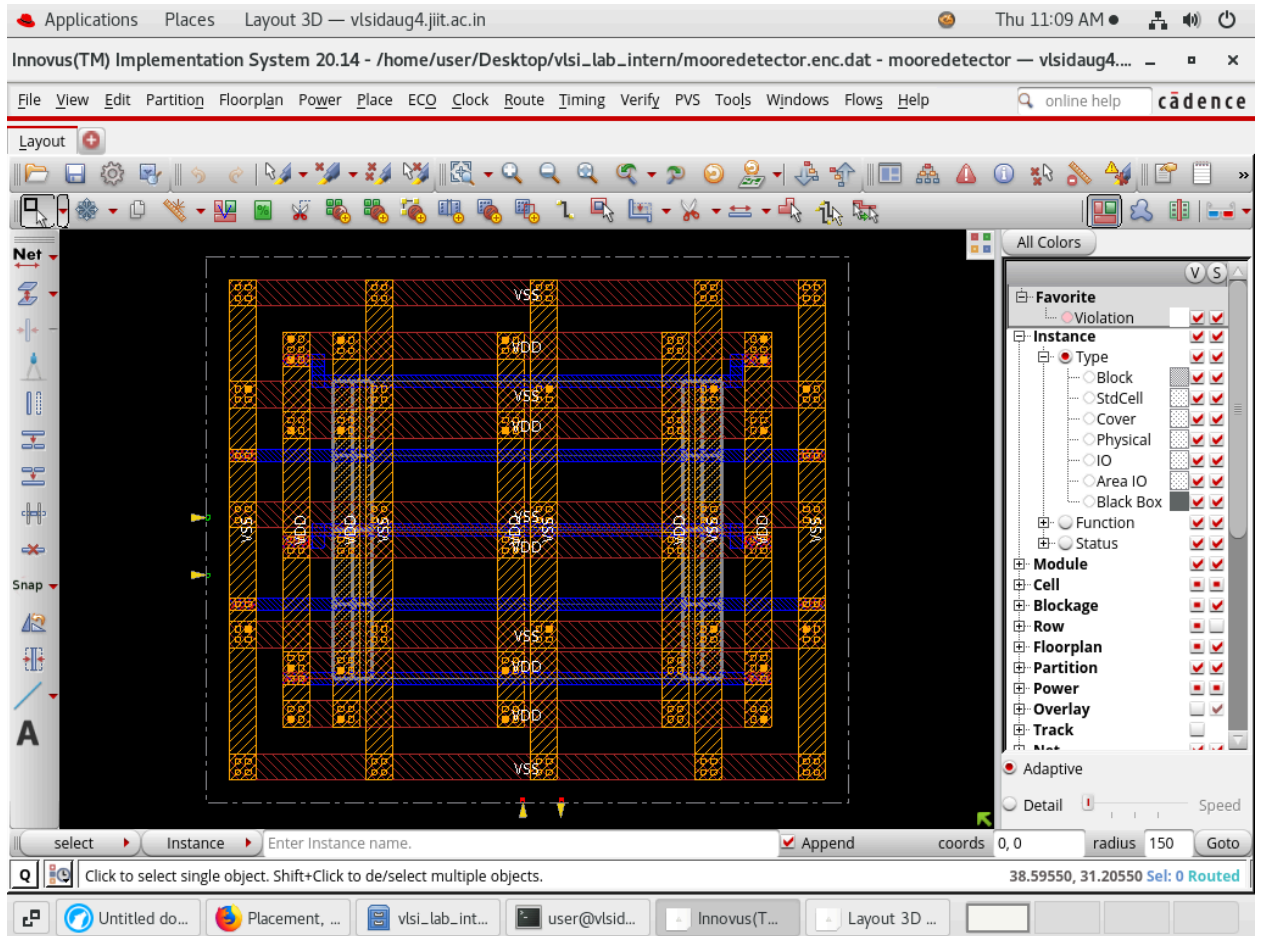
* Total instances in design with no power: 0

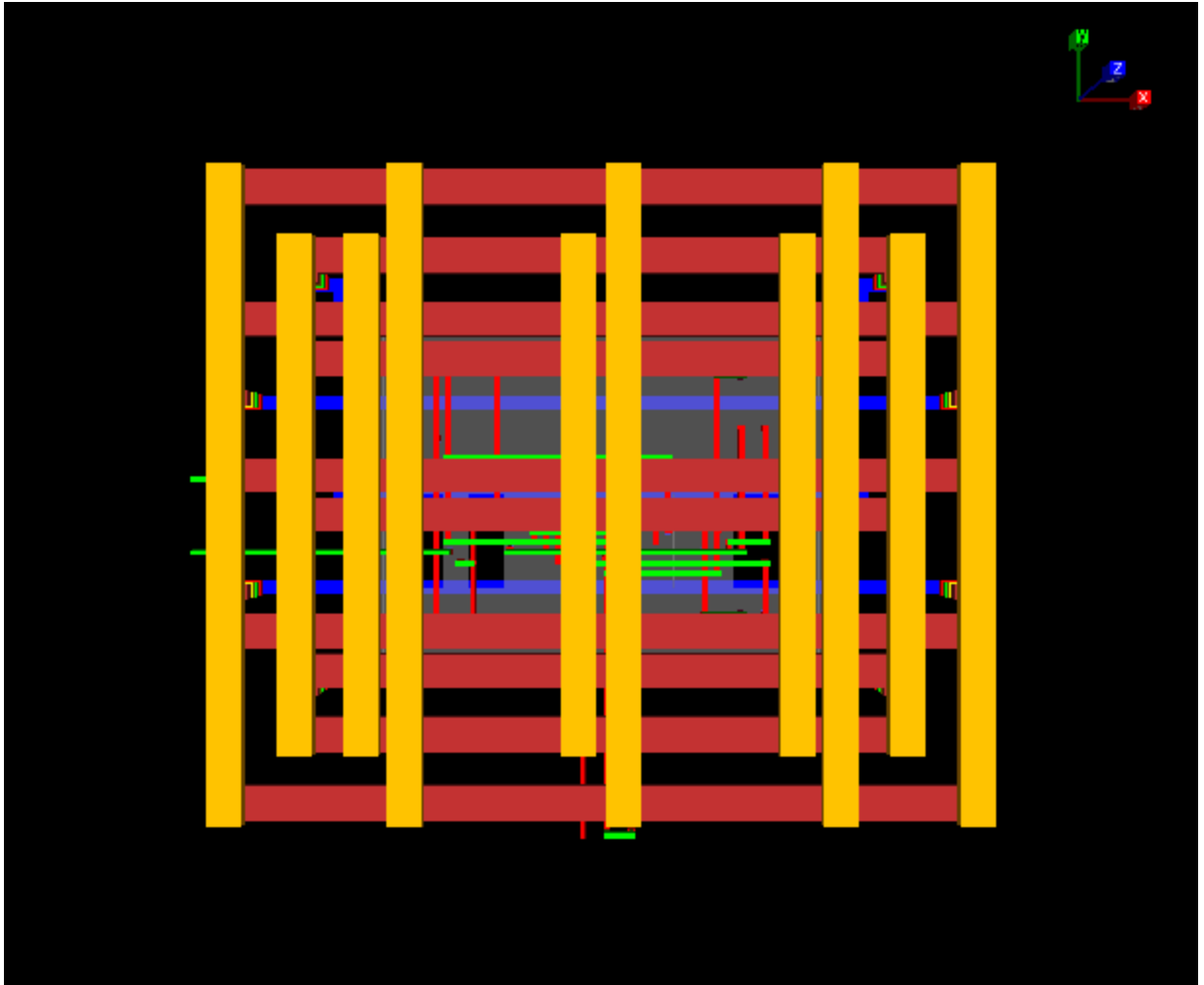
* Total instances in design with no activity: 0

* Total Fillers and Decap: 0

Ended Static Power Report Generation: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1172.05MB/2712.70MB/1172.10MB)

2. FSM - MOORE SEQUENCE DETECTOR





TIMING REPORT

Calculate delays in BcWc mode...

Start delay calculation (fullDC) (1 T). (MEM=1548.07)

Total number of fetched objects 19

AAE_INFO: Total number of nets for which stage creation was skipped for all views 0

End delay calculation. (MEM=1630.96 CPU=0:00:00.0 REAL=0:00:00.0)

End delay calculation (fullDC). (MEM=1603.89 CPU=0:00:00.0 REAL=0:00:00.0)

*** Done Building Timing Graph (cpu=0:00:00.1 real=0:00:00.0 totSessionCpu=0:02:40
mem=1603.9M)

timeDesign Summary

Setup views included:

WORST

+-----+-----+-----+-----+

Setup mode	all	reg2reg	default
WNS (ns):	0.000	N/A	0.000
TNS (ns):	0.000	N/A	0.000
Violating Paths:	0	N/A	0
All Paths:	0	N/A	0

	Real	Total
DRVs		
Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000
max_tran	0 (0)	0.000
max_fanout	0 (0)	0
max_length	0 (0)	0

Density: 86.719%

Routing Overflow: 0.00% H and 0.00% V

Reported timing to dir timingReports

Total CPU time: 0.1 sec

Total Real time: 0.0 sec

Total Memory Usage: 1571.148438 Mbytes

*** timeDesign #1 [finish] : cpu/real = 0:00:00.1/0:00:00.6 (0.2), totSession cpu/real = 0:02:39.6/0:25:13.0 (0.1), mem = 1571.1M

AREA REPORT

nnovus 1> report_area

Hinst Name	Module Name	Inst Count	Total Area
mooredetector		13	379.490

Power report

Power Net Detected:

Voltage	Name
0V	VSS
0.9V	VDD

Using Power View: WORST.

Set Default Frequency 100MHz.

Begin Power Analysis

0V	VSS
0.9V	VDD

Begin Processing Timing Library for Power Calculation

Begin Processing Timing Library for Power Calculation

Begin Processing Power Net/Grid for Power Calculation

Ended Processing Power Net/Grid for Power Calculation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=1193.47MB/2728.68MB/1429.41MB)

Begin Processing Timing Window Data for Power Calculation

Ended Processing Timing Window Data for Power Calculation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=1193.59MB/2728.68MB/1429.41MB)

Begin Processing User Attributes

Ended Processing User Attributes: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=1193.65MB/2728.68MB/1429.41MB)

Begin Processing Signal Activity

Starting Levelizing

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 10%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 20%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 30%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 40%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 50%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 60%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 70%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 80%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 90%

Finished Levelizing

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)

Starting Activity Propagation

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)

** INFO: (VOLTUS_POWR-1356): No default input activity has been set. Defaulting to 0.2.

Use 'set_default_switching_activity -input_activity' command to change the default activity value.

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 10%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 20%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 30%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 40%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 50%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 60%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 70%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 80%

Finished Activity Propagation

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)

Ended Processing Signal Activity: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1193.91MB/2728.68MB/1429.41MB)

Begin Power Computation

of cell(s) missing both power/leakage table: 0

of cell(s) missing power table: 0

of cell(s) missing leakage table: 0

of MSMV cell(s) missing power_level: 0

Starting Calculating power

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)

... Calculating switching power

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 10%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 20%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 30%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 40%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 50%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 60%

... Calculating internal and leakage power

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 70%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 80%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 90%

Finished Calculating power

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)

Ended Power Computation: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1194.35MB/2728.68MB/1429.41MB)

Begin Processing User Attributes

Ended Processing User Attributes: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1194.35MB/2728.68MB/1429.41MB)

Ended Power Analysis: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1194.43MB/2728.68MB/1429.41MB)

Begin Boundary Leakage Calculation

Ended Boundary Leakage Calculation: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1194.43MB/2728.68MB/1429.41MB)

Begin Static Power Report Generation

```
*-----
*   Innovus 20.14-s095_1 (64bit) 04/19/2021 14:41 (Linux 2.6.32-431.11.2.el6.x86_64)
*
*
*   Date & Time:   2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)
*
*-----
*
*   Design: mooredetector
*
*   Liberty Libraries used:
*       WORST:
* /home/user/Desktop/vlsi_lab_intern/mooredetector.enc.dat/libs/lib/typ/slow.lib
*
*   Parasitic Files used:
*
*   Power View : WORST
*
*   User-Defined Activity : N.A.
*
*   Activity File: N.A.
*
*   Hierarchical Global Activity: N.A.
*
```

* Global Activity: N.A.
 *
 * Sequential Element Activity: N.A.
 *
 * Primary Input Activity: 0.200000
 *
 * Default icg ratio: N.A.
 *
 * Global Comb ClockGate Ratio: N.A.
 *
 * Power Units = 1mW
 *
 * Time Units = 1e-09 secs
 *
 * report_power
 *

Total Power

Total Internal Power:	0.00085341	78.5743%
Total Switching Power:	0.00023040	21.2134%
Total Leakage Power:	0.00000231	0.2123%
Total Power:	0.00108612	

Group	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
Sequential	0.0006012	5.512e-05	1.683e-06	0.000658	60.58
Macro	0	0	0	0	
IO	0	0	0	0	
Combinational	0.0002522	0.0001753	6.223e-07	0.0004281	39.42
Clock (Combinational)		0	0	0	
Clock (Sequential)		0	0	0	
Total	0.0008534	0.0002304	2.306e-06	0.001086	100

Rail	Voltage	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
VDD	0.9	0.0006767	0.0001816	1.935e-06	0.0008602	79.2
Default	0.9	0.0001767	0	3.706e-07	0.0001771	16.3

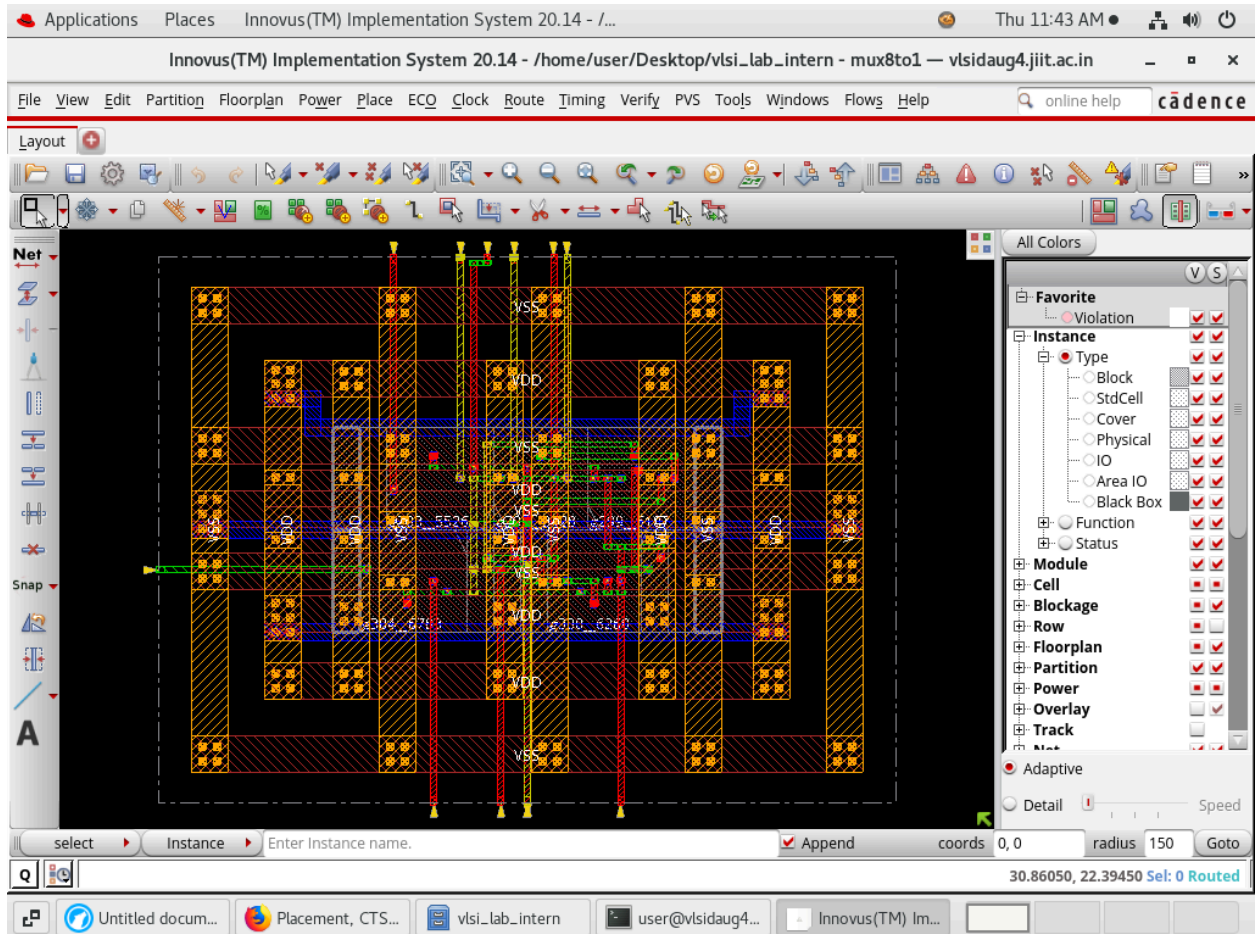
```
* Power Distribution Summary:
*   Highest Average Power:      state_reg[0] (DFFRX4): 0.0002583
*   Highest Leakage Power:      state_reg[1] (DFFRX4): 5.611e-07
*   Total Cap: 4.49202e-14 F
*   Total instances in design: 13
*   Total instances in design with no power: 0
*   Total instances in design with no activity: 0

*   Total Fillers and Decap: 0
```

Ended Static Power Report Generation: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1194.70MB/2728.68MB/1429.41MB)

1

3. Combinational MUX(8:1)



TIMING REPORT

timeDesign Summary

Setup views included:

WORST

Setup mode	all	default
WNS (ns):	0.000	0.000
TNS (ns):	0.000	0.000
Violating Paths:	0	0
All Paths:	0	0

Real	Total

DRVs	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	0 (0)
max_fanout	0 (0)	0	0 (0)
max_length	0 (0)	0	0 (0)

Density: 92.000%

Routing Overflow: 0.00% H and 0.00% V

Reported timing to dir timingReports

Total CPU time: 0.13 sec

Total Real time: 1.0 sec

Total Memory Usage: 1511.808594 Mbytes

*** timeDesign #1 [finish] : cpu/real = 0:00:00.1/0:00:00.7 (0.2), totSession cpu/real = 0:01:07.2/0:08:52.9 (0.1), mem = 1511.8M

AREA REPORT

Hinst Name	Module Name	Inst Count	Total Area
mux8to1		11	157.802

Power report

Power Net Detected:

Voltage	Name
0V	VDD
0.9V	VSS

Using Power View: WORST.

Set Default Frequency 100MHz.

Begin Power Analysis

0V	VDD
0.9V	VSS

Warning:

There are 2 power/gnd nets that are not connected

VDD VSS ...

Use 'globalNetConnect' to define rail connections.

** WARN: (VOLTUS_POWR-2035): There are 2 power/gnd nets that are not connected

VDD VSS ...

Use 'globalNetConnect' to define rail connections.

Begin Processing Timing Library for Power Calculation

Begin Processing Timing Library for Power Calculation

Begin Processing Power Net/Grid for Power Calculation

Ended Processing Power Net/Grid for Power Calculation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=1136.52MB/2672.97MB/1323.70MB)

Begin Processing Timing Window Data for Power Calculation

Ended Processing Timing Window Data for Power Calculation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=1136.64MB/2672.97MB/1323.70MB)

Begin Processing User Attributes

Ended Processing User Attributes: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=1136.70MB/2672.97MB/1323.70MB)

Begin Processing Signal Activity

Starting Levelizing

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT)

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 10%

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 20%

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 30%

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 40%

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 50%

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 60%

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 70%

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 80%

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 90%

Finished Levelizing

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT)

Starting Activity Propagation

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT)

** INFO: (VOLTUS_POWR-1356): No default input activity has been set. Defaulting to 0.2.

Use 'set_default_switching_activity -input_activity' command to change the default activity value.

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 10%
2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 20%
2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 30%
2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 40%
2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 50%

Finished Activity Propagation

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT)
Ended Processing Signal Activity: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1136.93MB/2672.97MB/1323.70MB)

Begin Power Computation

```
-----  
# of cell(s) missing both power/leakage table: 0  
# of cell(s) missing power table: 0  
# of cell(s) missing leakage table: 0  
# of MSMV cell(s) missing power_level: 0  
-----
```

Starting Calculating power

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT)
... Calculating switching power
2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 10%
2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 20%
2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 30%
Cannot locate supply power rail for net 'y' of instance g298__2398
2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 40%
Cannot locate supply power rail for net 'n_0' of instance g308
Cannot locate supply power rail for net 'n_1' of instance g305
Cannot locate supply power rail for net 'n_2' of instance g304__6783
Cannot locate supply power rail for net 'n_3' of instance g303__5526
only first five unconnected nets are listed...
2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 50%
2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 60%
... Calculating internal and leakage power
2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 70%

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 80%

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT): 90%

Finished Calculating power

2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT)

Ended Power Computation: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1137.38MB/2672.97MB/1323.70MB)

Begin Processing User Attributes

Ended Processing User Attributes: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1137.38MB/2672.97MB/1323.70MB)

Ended Power Analysis: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1137.46MB/2672.97MB/1323.70MB)

Begin Boundary Leakage Calculation

Ended Boundary Leakage Calculation: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1137.46MB/2672.97MB/1323.70MB)

Begin Static Power Report Generation

```
*-----  
*   Innovus 20.14-s095_1 (64bit) 04/19/2021 14:41 (Linux 2.6.32-431.11.2.el6.x86_64)  
*  
*  
*   Date & Time:   2025-Jun-26 11:47:46 (2025-Jun-26 06:17:46 GMT)  
*  
*-----  
*  
*   Design: mux8to1  
*  
*   Liberty Libraries used:  
*       WORST: genus/library/slow.lib  
*  
*   Parasitic Files used:  
*  
*   Power View : WORST  
*  
*   User-Defined Activity : N.A.  
*  
*   Activity File: N.A.  
*  
*   Hierarchical Global Activity: N.A.  
*  
*   Global Activity: N.A.
```

```

*
* Sequential Element Activity: N.A.
*
* Primary Input Activity: 0.200000
*
* Default icg ratio: N.A.
*
* Global Comb ClockGate Ratio: N.A.
*
* Power Units = 1mW
*
* Time Units = 1e-09 secs
*
* report_power
*

```

```

-----
*

```

Total Power

```

-----
Total Internal Power:    0.00060981    75.7936%
Total Switching Power:   0.00019399    24.1115%
Total Leakage Power:     0.00000076    0.0949%
Total Power:             0.00080456
-----

```

Group	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
Sequential	0	0	0	0	0
Macro	0	0	0	0	0
IO	0	0	0	0	0
Combinational	0.0006098	0.000194	7.638e-07	0.0008046	100
Clock (Combinational)	0	0	0	0	0
Clock (Sequential)	0	0	0	0	0
Total	0.0006098	0.000194	7.638e-07	0.0008046	100

Rail	Voltage	Internal	Switching	Leakage	Total	Percentage
------	---------	----------	-----------	---------	-------	------------

	Power	Power	Power	Power (%)	
Default	0.9	0.0006098	0.000194	7.638e-07	0.0008046 100

* Power Distribution Summary:

* Highest Average Power: g304__6783 (MXI2X2): 0.0001316

* Highest Leakage Power: g303__5526 (MXI2X2): 1.552e-07

* Total Cap: 1.86084e-14 F

* Total instances in design: 11

* Total instances in design with no power: 0

* Total instances in design with no activity: 0

* Total Fillers and Decap: 0

Ended Static Power Report Generation: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1137.73MB/2672.97MB/1323.70MB)